

MAT 1001 Midterm Exam, 9:00-11:00am, October 25, 2025

Your Name and Student ID:

Your Lecture Class(e.g, L01):

Instruction: (i) This is a closed-book and closed-notes exam; no calculators, no dictionaries and no cell phones, no cheat sheet; (ii) Show your work unless otherwise instructed—a correct answer without showing your work when required shall be given no credits; (iii) Write down ALL your work and your answers(including the answers for short questions) in the Exam Book; (iv) Make sure that your exam paper has 4 pages; if not, report it to the proctor in the exam room; and (v) The full score for the exam is 174 points, to earn all of which you need to answer ALL questions correctly.

1. (45 points) Short Questions (for these questions, NO need to show your work, just write down your answers in the Exam Book; NO partial credits for each question)

(i). Given $f(t) = \cos t$, $f^{(27)}(t)$ is equal to

- (a) $\sin t$.
- (b) $-\sin t$.
- (c) $\cos t$.
- (d) $-\cos t$.

(ii). Consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where

$$P(x) = a_n x^n + \dots + a_1 x + a_0$$

$$Q(x) = b_k x^k + \dots + b_1 x + b_0$$

with $a_n > 0$ and $b_k > 0$. Consider the limit of $f(x)$ as $x \rightarrow +\infty$ or $x \rightarrow -\infty$, and let's allow the limit to be either $+\infty$ or $-\infty$.

- (a) If $n = k$, then $\lim_{x \rightarrow +\infty} f(x) = -\lim_{x \rightarrow -\infty} f(x)$.
- (b) If $n > k$, then $\lim_{x \rightarrow +\infty} f(x) = 0$.
- (c) If $n > k$, then $\lim_{x \rightarrow -\infty} f(x) = -\infty$.
- (d) None of the above.

(iii). Find all possible asymptotes (horizontal, vertical and oblique) of $f(x) = \frac{x^2+1}{x-1}$.

(iv). True or False: The equation $x^3 - 3x + 1 = 0$ has at least two roots in $[0, 2]$.

True

False

(v). True or False: Suppose $(f(x))^3$ is continuous, then $f(x)$ is also continuous.

True

False

(vi). Find an equation of the tangent line to $f(x) = \tan x + 9 \cos x$ at $x = \pi$:

(vii). True or False: If $y = f(x)$ is differentiable at x_0 , then $f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + \varepsilon\Delta x$ (or in another form, $= f'(x_0)\Delta x + o(\Delta x)$) for some $\varepsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

True

False

(viii). Let $g(x) = |x|$ on $[-1, 1]$. Which is correct?

(a) There exists at least one $c \in (-1, 1)$ with $g'(c) = 0$ by Rolle's Theorem.

(b) There exists exactly one $c \in (-1, 1)$ with $g'(c) = 0$ by Rolle's Theorem.

(c) Rolle's Theorem does not apply because g is not continuous at 0.

(d) Rolle's Theorem does not apply because g is not differentiable on $(-1, 1)$.

(ix). True or False: Suppose $f(x)$ has an inflection point $(c, f(c))$. Then near $(c, f(c))$ the graph of $f(x)$ crosses the tangent line passing through $(c, f(c))$.

True

False

(x). True or False: If $f''(x)$ exists everywhere on $(-\infty, +\infty)$, and $f(0)$ is a local minimum, then 0 is a critical point and $f''(0) > 0$.

True

False

(xi). If $a = \int_{-\pi/2}^{\pi/2} \frac{\sin x}{1+x^2} \cos^4 x dx$, $b = \int_{-\pi/2}^{\pi/2} (\sin^3 x + \cos^4 x) dx$, and $c = \int_{-\pi/2}^{\pi/2} (x^2 \sin^3 x - \cos^4 x) dx$, then

(a) $b < c < a$.

(b) $a < c < b$.

(c) $b < a < c$.

(d) $c < a < b$.

(xii). Let the function $f(x)$ be positive and continuous on $[a, b]$. Then the number of roots of the equation $\int_a^x f(t)dt + \int_b^x f(t)dt = 0$ in (a, b) is

(a) 0.

(b) 1.

(c) 2.

- (d) 3.
- (xiii) . Use Newton's method to approximate the root of the equation $\cos x = 2x$ with the initial guess $x_0 = 0$, then the next approximation of the root is _____.
- (xiv). Use the definition of definite integrals and a special case of Riemann sum to express the limit $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right)$ as a definite integral (Hint: $\frac{1}{n+i} = \frac{1}{n(1+i/n)}$). Do not try to calculate the value of the definite integral. _____
- (xv). Find the indefinite integral $\int (2x(1 - x^{-3}) - \sec^2(2x)) dx$ _____.

2. (24 points) Find each of the following limits or explain why the limit does not exist (the limit is allowed to be ∞ or $-\infty$). *You are NOT allowed to use L'Hopital's Rule—no credits if you use it!*

(i) $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\tan(2x)}$.

(ii) $\lim_{x \rightarrow \pi} \frac{\sin(3x)}{\tan(2x)}$.

(iii) $\lim_{x \rightarrow 2} f(x)$, where

$$f(x) = \begin{cases} (x-2)^4 \sin\left(\frac{1}{x-2}\right), & \text{if } x < 2; \\ x^2 - x - 6, & \text{if } x \geq 2. \end{cases}$$

(iv) $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 - x})$

3. (24 points) Find the derivatives of the following functions or explain why the derivative does not exist.

(i) Consider the function

$$f(x) = \begin{cases} (x-1)^2 \sin\left(\frac{1}{x-1}\right), & x \neq 1 \\ 1, & x = 1. \end{cases}$$

Find $f'(1)$.

(ii) Find $f'(2)$, where $f(x)$ is defined in part (i) above.

(iii) Let $y^2 - 2x = 1 - 2y$. Find $\frac{dy}{dx}$ (express your answer in terms of x and y).

(iv) Find $\frac{d^2y}{dx^2}$, where y is defined implicitly in part (iii) above (express your answer in terms of x and y).

4. (12 points) Suppose a particle moves on the s -axis, with acceleration function being $a(t) = \cos t$, $t \geq 0$, and initial velocity $v(0) = 1/2$ and initial position $s(0) = 2$.

- (i) Find the position function $s(t)$.
 - (ii) Find the total distance travelled by the particle from $t = 0$ to $t = 2\pi$.
5. (20 points) Suppose the cost function for producing x TVs in a factory is given by $C(x) = 20000 + 150x + 0.5x^2$ for $x > 0$.
- (i) (4 points) Write down the average cost function $a(x)(= C(x)/x)$ and the marginal cost function $m(x)$.
 - (ii) (6 points) Find x that minimize the average cost.
 - (iii) (4 points) Find the intersection point of the average cost curve and the marginal cost curve.
 - (iv) (6 points) Prove that for a **general** differentiable cost function, when the average cost function reaches its minimum, the average cost must equal to the marginal cost.
6. (8 points) A spherical balloon is inflated with gas at the rate 20π m³/min. How fast is the balloon's radius increasing at the moment when the radius is 1 m? Recall that the volume formula for a sphere is $V = 4\pi r^3/3$.
7. (25 points) Let $f(x) = (x - 2)^{2/3} - (x - 2)$ on $[0, 4]$.
- (i) (4 points) Find all critical points in $[0, 4]$.
 - (ii) (6 points) Determine the intervals of increase/decrease.
 - (iii) (3 points) Determine all locations (that is, the x -coordinates) of local maxima (possibly including endpoints).
 - (iv) (3 points) Compute the global/absolute maximum value of f on $[0, 4]$ and where it occurs.
 - (v) (6 points) Find the intervals on which $f(x)$ is concave up or concave down.
 - (vi) (3 points) Does $f(x)$ have an inflection point?
8. (8 points) Use Mean Value Theorem to prove that for $x \in (0, \pi/2)$, we have $\sin x < x$ and $\tan x > x$.
9. (8 points) John, a CUHKSZ freshman, is 3 km offshore in a boat and wishes to reach Yantian Port 5 km down a straight shoreline from the point nearest the boat. He can row 3 km/h and can walk 6 km/h. Where should he land his boat to reach the port in the least amount of time?